

CHAOTIC DYNAMICS OF SEA CLUTTER: AN EXPERIMENTAL STUDY

Simon Haykin and Sadasivan Puthusserypady

Communications Research Laboratory, McMaster University, Canada.

ABSTRACT

In this paper, we demonstrate, convincingly for the first time, the chaotic dynamics of sea clutter. This is based on the results obtained from a series of extensive and thorough experiments, which were made with different sea clutter data sets. The study includes correlation dimension analysis (based on the maximum likelihood principle) and Lyapunov spectrum analysis. The Lyapunov (Kaplan-Yorke) dimension, which is a by-product of the Lyapunov spectrum analysis, is found to be a good estimator of the correlation (fractal) dimension.

INTRODUCTION

Chaos defines the highly complex/irregular behavior of a low-dimensional dynamical system, which is both deterministic and nonlinear. This represents a powerful notion, permitting the use of a simple deterministic nonlinear system to explain highly irregular behavior exhibited by various physical phenomena in nature. Since Lorenz's classic work in 1963, the theory of chaos has been applied to many physical systems to model the system as a chaotic process, which has otherwise been modeled as a stochastic process. Similar studies have been extended to the analysis of sea clutter (radar backscatter from sea surface) and the results suggest that there may be a low-dimensional dynamical attractor controlling sea clutter.

Haykin and Leung were the first to claim the chaotic dynamics of sea clutter by looking at the (fractal) correlation dimension (1,2). They used the Grassberger-Procaccia algorithm to estimate the correlation dimension and its value was reported to be between 6 and 9. In 1992, He and Haykin moved one more step forward to show the chaotic dynamics of sea clutter by calculating the largest Lyapunov exponent (3). They showed that the largest Lyapunov exponent is positive (sure test for the existence of chaotic dynamics). Later, in 1995, Haykin and Li expanded further on the chaotic dynamics of sea clutter, and they made use of this property of sea clutter for the enhanced detection of a target in the presence of sea clutter (4). Studies by Palmer *et al* have also shown that sea clutter can be modeled as a low-dimensional attractor, and they reported a correlation dimension between 5 and 8 (5). Our experimental results reported herein

convincingly demonstrate the chaotic dynamics of sea clutter. This means that the underlying dynamics of sea clutter can be described by a completely deterministic model showing chaotic behavior. The state of the system can be described by a set of variables and its evolution is given by a set of coupled nonlinear differential/difference equations. We have made extensive study of the chaotic dynamics of sea clutter (under various conditions) by calculating the correlation dimension and the Lyapunov spectrum (along with the Kaplan-Yorke dimension), and the results conclusively demonstrate that sea clutter is indeed chaotic.

RADAR AND DATABASE

The database used for the present study was collected by using an instrument quality radar called the IPIX radar. The IPIX radar has the following features. It is dual-polarized and coherent. It has built-in calibration equipment, which permits quantitative measurements to be made with high confidence, and a digital control system which allows the experimental parameters to be easily varied.

Experiments at Cape Bonavista, Newfoundland: The IPIX radar system was first installed at Cape Bonavista, Newfoundland, Canada, in May 1989. A wide variety of sea states can be observed at that site throughout the year. The radar antenna was mounted at a height of about 25m above the water, similar to that of a typical ship-mounted radar. During the field trip of June 1989, sea clutter for various sea states were collected. For each data package, there are I-channel data and Q-channel data, each consisting of at least 50,000 samples from consecutive sweeps.

Experiments at Dartmouth, Nova Scotia: Further field experiments at Dartmouth, Nova Scotia, Canada, were carried out in November 1993, using a refined version of the IPIX radar. The radar antenna was mounted about 30m above the water level and within 15m of the shore.

Experiments at Argentia, Newfoundland: For these experiments, an incoherent commercial radar system was installed at the VTS site in Argentia, Newfoundland, in January 1994. This produced

only the amplitude, different from the in-phase and quadrature components in the other experiments.

Pre-processing of the Collected Data

The experimental data (radar data) thus collected from different sites using different radars were influenced by many types of errors. These errors were minimized/removed by suitable pre-processing of the data before further analysis of the data for extracting the chaotic parameters. At the same time, the pre-processing was performed in such a way that it would not affect the dynamic characteristics of the underlying system. In this study, the following pre-processing was applied to the radar data.

Amplitude and phase corrections. The in-phase and quadrature components of the coherent radar showed amplitude and phase mismatch (unequal variance and nonzero cross correlation) because of experimental errors. Appropriate corrections were incorporated in the in-phase and quadrature components. The corrected in-phase and quadrature components were orthogonal and showed equal variance, as desired.

Filtering. The collected data may be influenced by measurement noise or additive noise. These artifacts are to be attenuated/minimized by applying suitable filters to the recorded signal. Care must be taken in choosing the smoothing filters for a chaotic time series, because the filtering may affect the underlying dynamics of the system which produced the time series (6). We used two kinds of filtering to reduce the effect of experimental noise. In the first method, we used a smoothing filter of the form $\hat{x}(n) = \sum_{i=1}^3 a_i x(n+i-2)$ with $a_i = 1/3$ to smooth the raw clutter data. In the second method, we designed a low-pass linear phase finite impulse response (FIR) filter having 100 coefficients to filter the raw clutter data. While designing the filter, we made sure that the bandwidth of the filter was large enough so as not to attenuate the clutter data.

EXTRACTION OF CHAOTIC INVARIANTS FROM THE EXPERIMENTAL TIME SERIES

Characterizing the dynamical systems that one observes is a critical part of the analysis of measured signals. Two important characteristic parameters of a chaotic process are (1) **correlation (fractal) dimension** (D_2), and (2) **Lyapunov exponents** (λ_i) and hence the **Lyapunov dimension**.

Phase-space reconstruction. The construction of a phase-space or embedding space is an important step in nonlinear analysis of experimental time series where the unidimensional data are transformed into a multidimensional space. The experimental time se-

ries is a single-dimensional data set where it has only one phase variable at any instant of time. In the late 1800's, Poincaré showed that much can be learnt about the dynamical behavior of the system from the analysis of its trajectories in a multidimensional state-space in which a single point characterizes the system at an instant of time. The experimentalist's dilemma has been that for a system with d_E degrees of freedom, it seems necessary to measure d_E independent variables, an almost impossible chore for a complex system. This problem found a solution in the embedding theorem of Takens, which states that a multidimensional state-space can be constructed from the measurement of a single variable (7). That is, for a time series $s(nT)$, $n = 0, 1, \dots, N$, (where N is the total number of samples and T is the sampling time), the state-space vector $s(n)$ is constructed by assigning co-ordinates

$$\begin{aligned} s_1(n) &= s(n) \\ s_2(n) &= s(n + \tau) \\ &\vdots \\ s_{d_E}(n) &= s(n + (d_E - 1)\tau) \end{aligned} \quad (1)$$

where d_E is the embedding dimension and τ is the characteristic time delay. These parameters are not chosen arbitrarily; rather, they are determined experimentally using the *False Nearest Neighbor* analysis and *Mutual Information*, respectively (8).

Test of stationarity (*Recurrence plots*). The dynamical analysis of experimental time series requires stationary data sets. Ordinarily, one assumes stationarity of the phenomenon under consideration. Therefore, the first step in every analysis must be to check for stationarity of the time series. Eckmann and Ruelle proposed an elegant graphical tool, called the "recurrence plot" for diagnoses of the drift and hidden periodicities in the time evolution of dynamical systems, which are unnoticeable otherwise (9).

Tests of nonlinearity. The nonlinearity of clutter data was tested in three different ways. In the first method, we used the method of surrogate data analysis. The growth of the inter-point distances as the discriminating statistic (for 50 surrogate data sets), with an average Mann-Whitney rank-sum statistics (Z-value) < -3.0 , rejects the null hypothesis that each clutter sample is that of a linear process. A Z-value of less than -3.0 is considered ground for strong rejection (10).

In the second method, we applied the surrogate data analysis proposed by Theiler *et al.*, to several clutter data sets with correlation dimension as the discriminating statistic (11). The results on correlation dimension analysis of the original and the corresponding surrogate data sets showed that they are statistically distinct, justifying rejection of the null hypothe-

sis that each clutter sample is that of a linear stochastic process.

In the third method (which gives the most convincing proof that sea clutter is indeed a nonlinear process), we used a wave-shaping filter stimulated by pure white Gaussian noise to produce a signal (with the same power spectrum as that of a typical clutter data). Tests showed that this signal has statistically distinct chaotic parameters from those of the original clutter data. The chaotic analysis inferred it as noise rather than a chaotic process (with very many positive exponents, and increased non-converging correlation dimension with increased embedding dimension).

Correlation Dimension (D_2)

The correlation dimension has been the most intensely studied characteristic invariant for dynamical systems. Unlike the integer Euclidean dimension, the correlation dimension is fractional in nature. This dimension of a dynamical system directly reflects the way in which points on the attractor are distributed in the phase-space or embedding space. D_2 reflects the complexity of the underlying dynamical system and measures the degrees of freedom or the number of parameters required to describe the system under investigation. The correlation dimension is estimated from the measured time series by looking at the way in which points within a sphere of radius r scales as the radius shrinks to zero in the embedding space. Grassberger and Procaccia developed an efficient algorithm for computing the correlation dimension based on the above idea (12). Unfortunately, this algorithm suffers from certain disadvantages, such as spurious estimates of D_2 in the presence of measurement noise. Schouten *et al* have come up with a more efficient method (based on the maximum likelihood principle) for estimating the correlation dimension in the presence of noise (13). In this study, we used this later method for the estimation of correlation dimension of sea clutter data.

Lyapunov Exponents (λ_i)

The Lyapunov spectrum (i.e., set of Lyapunov exponents) of dynamical systems is one of the important invariants that characterize the attractors of a dynamical system. They are independent of initial conditions on any orbit and are therefore fundamental properties of the attractor geometry and the dynamics. The Lyapunov exponents may be computed by a recursive QR-decomposition technique from the Jacobian of a function, $\mathbf{F}[\mathbf{s}(n)]$, which maps points on the orbit into points at t time steps later (8,14):

$$\mathbf{s}(n+t) = \mathbf{F}[\mathbf{s}(n)]. \quad (2)$$

The terms in the Jacobian are found by a least-squares algorithm (8,14). The Lyapunov exponents are then calculated as the logarithm of the eigenval-

ues of products of such Jacobian matrices for a number of steps along the trajectory of the orbit (8,14).

Lyapunov Dimension or Kaplan-Yorke Dimension (D_{KY})

Once the Lyapunov characterizing exponents are obtained, we can compute the Lyapunov dimension or Kaplan-Yorke dimension D_{KY} using the following formula (12):

$$D_{KY} = K + \frac{\sum_{a=1}^K \lambda_a}{|\lambda_{K+1}|} \quad (3)$$

where the exponents are ordered in descending order $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_{d_E}$ and K is the largest integer for which $\lambda_1 + \lambda_2 + \dots + \lambda_K \geq 0$.

RESULTS AND DISCUSSION

In this section, we discuss the results of our experimental studies on sea clutter data. The first sub-section discusses the correlation dimension analysis and the second sub-section discusses the Lyapunov spectrum analysis.

Each sea clutter data set has at least 50,000 samples. In each case, the data were pre-processed (as explained previously) for amplitude and phase correction (coherent radar data) and for reducing the effect of noise (coherent as well as incoherent radar data). After the preprocessing, each data set was analyzed to test whether the pre-processing had altered the underlying dynamics (6). This was done statistically by analyzing the *continuity* and *differentiability* between the original and pre-processed data embeddings. In all the cases, the filtered data passed the test, meaning that the dynamics remained unchanged after the pre-processing. After this, the data were subjected to tests for stationarity and nonlinearity (9,10,11). The test results consistently showed that the data were stationary over a sufficient length of time, and nonlinear. This enabled us to continue the further analysis to extract the chaotic parameters and to characterize the sea clutter as a chaotic process.

Correlation Dimension

As indicated earlier, correlation dimension is the most basic and static property of an attractor. We have used the software (RRCHAOS) developed by the Chemical Reactor Engineering section of Delft University for the estimation of the correlation dimension (10). The maximum-likelihood method of Schouten *et al.*, was implemented in software for the estimation of D_2 (13).

In this work, we made an extensive study of the D_2 of sea clutter. We used data from different radar sites with varying radar parameters, sea states, and

ranges. For the coherent radar case, the data analyzed include the amplitude and phase corrected and unfiltered in-phase and quadrature components, and its amplitudes, the 3-point smoothed and the FIR filtered in-phase and quadrature components and its amplitude. For the incoherent radar case, it includes the unfiltered amplitude and 3-point smoothed and FIR filtered amplitudes. The multidimensional embedding parameters (embedding dimension and time delay) of the single-dimensional sea clutter time series were obtained by appropriate methods (8). The maximum-likelihood estimates of the D_2 values for different data showed that the value varies between 4.1 and 4.5, irrespective of changes in the environmental and radar parameters. This indeed shows that sea clutter has a strange attractor with complexity lying between 4.1 and 4.5. The D_2 values for in-phase, quadrature and amplitude components remain consistent for all data sets, which again proves the delay embedding theorem of Takens for sea clutter data (7).

Lyapunov Spectrum

The determination of the Lyapunov exponents is the single most important measurement in the analysis of a possible chaotic system. This is because the Lyapunov exponents not only tell us about sensitive dependence on the initial conditions of nearby trajectories but also tell us how orbits on the attractor move apart/together under the evolution of the dynamics. If the exponents are all zero or negative, the trajectories do not diverge and the system is stable and not chaotic. If one or more Lyapunov exponents are positive, the trajectories diverge, the system is unstable, and the source of the signal is executing chaotic motion. For this part of the study, we employed the software developed by Del Mar (Chaotic Signal Processing) for computing all the Lyapunov exponents (equal to the embedding dimension) of sea clutter (15). The algorithm gave 5 or 6 exponents. The important observations are as follows:

- The largest two exponents are always positive.
- The third exponent goes to zero, consistently indicating that sea clutter is the result of $d_E = 5$ or 6 coupled nonlinear differential equations, where d_E is the embedding dimension.
- The sum of all exponents is negative, indicating a dissipative dynamical, and hence a physically realizable system (as it should be).
- The magnitude of the positive exponents increases with increasing wave-heights (i.e., sea state).

Table.1 summarizes the results of Lyapunov spectrum analysis of one set of sea clutter data for different radar components and different ranges.

Lyapunov Dimension

The Lyapunov dimension or Kaplan-Yorke dimension for sea clutter was estimated from the Lyapunov exponents. The results show that it indeed matches very well with the estimated correlation dimension, which is again another strong evidence for the chaotic dynamics of sea clutter.

One other essential characteristic of a chaotic process is its short-term predictability, which, in turn, shows the deterministic nature of the process. It is short-term predictable because of the presence of positive Lyapunov exponents (recall the inverse relationship between the horizon of predictability and largest positive Lyapunov exponent). In separate studies using regularized radial-basis function (RBF) networks and support vector machines (SVM), we have successfully built neural network models for the dynamic reconstruction of sea clutter, given a time series made up of samples of the in-phase, quadrature-phase, or amplitude components.

CONCLUSION

The thorough and extensive experimental study reported in this paper has clearly demonstrated the chaotic dynamics of sea clutter in the most complete manner. This conclusion is based on the results from different nonlinearity tests and different analyses such as correlation dimension, Lyapunov exponents, Lyapunov dimension. Irrespective of location and range, the correlation dimension of sea clutter data varies between 4.1 and 4.5 (fractal) and it matches well with the Kaplan-Yorke dimension. The in-phase, quadrature, and amplitude components of all sea clutter data showed at least 2 positive Lyapunov exponents, with the third exponent essentially equal to zero invariably. With all these convincing results, coupled with successful dynamic reconstructions, we conclude that the underlying dynamics of sea clutter are indeed chaotic.

REFERENCES

1. Leung H and Haykin S., 1990, "Is there a radar clutter attractor," *Appl. Phys. Lett.*, **56**, 593-595.
2. Haykin S and Leung H., 1992, "Model reconstruction of chaotic dynamics: First preliminary radar results," *Proc. ICASSP*, **IV**, 125-128.
3. He N and Haykin S., 1992, "Chaotic Modeling of Sea Clutter," *Electronics Lett.*, **28**, 2076-2077.
4. Haykin S and B.Li X., 1995, "Detection of Signals in Chaos," *Proc. IEEE*, **83**, 94-122.
5. Palmer A.J, Kropfli R.A and Fairall C.W., 1995, "Signature of deterministic chaos in radar sea clutter and ocean surface winds," *Chaos*, **5**, 613-616.

6. Pecora L.M and Carrol T.L., 1996, "Discontinuous and non-differentiable functions and dimension increase, induced by filtering chaotic data," *Chaos*, **6**, 432-439.
7. Takens F, 1981, "Detecting strange attractors in turbulence," *Lecture Notes in Mathematics*, **898**, 366-381.
8. Abarbanel H.D.I, 1996, "Analysis of Observed Chaotic Data" Springer Verlag, New-York, Inc.
9. Eckmann J.P, Kamphorst S.O and Ruelle D., 1987, "Recurrence plots of dynamical systems," *Europhys. Lett.*, **4**, 973-977.
10. Schouten J.C and Van den Bleek C.M., 1994, *RRCHAOS Time Series analysis Software*, Section Chemical Reactor Engineering, Delft University of Technology, Delft, Netherlands.
11. Theiler J, Eubank S, Longtin A, Galdrikian B and Farmer J.D., 1992, "Testing for nonlinearity in time series: the method of surrogate data," *Physica D*, **58**, 77-94.
12. Grassberger P and Procaccia I., 1983, "Measuring the strangeness of strange attractors," *Physica D*, **9**, 189-208.
13. Schouten J C, Takens F and Van den Bleek C.M., 1994, "Estimation of dimension of a noisy attractor", *Phys. Rev. E*, **50**, 1851-1861.
14. Brown R, Bryant P and Abarbanel H.D.I., 1991, "Computing the Lyapunov exponents of a dynamical systems from observed time series," *Phys. Rev. A*, **43**, 2787-2806.
15. Tools for Dynamics (Chaotic Signal Processing (CSP) software), Applied Chaos, LLC, Del Mar, CA 92014-2620, 1995.

Table.1 Lyapunov Exponents of In-phase, Quadrature-phase and Amplitude of Clutter Data for Different Ranges (Dartmouth).

Lyapunov Exponents									
Range(m)	Amplitude			In-Phase Component			Quadrature Component		
	Corrected	3 Point	FIR	Corrected	3 Point	FIR	Corrected	3 Point	FIR
1545	+0.1998	+0.2325	+0.2693	+0.2730	+0.2692	+0.2929	+0.1888	+0.1728	+0.2422
	+0.0938	+0.1060	+0.1458	+0.1557	+0.1276	+0.1423	+0.0883	+0.0646	+0.1100
	-0.0207	-0.0260	+0.0183	+0.0353	+0.0025	+0.0208	-0.0111	-0.0089	+0.0018
	-0.1658	-0.2017	-0.1603	-0.1248	-0.1537	-0.1292	-0.1445	-0.1237	-0.1193
	-0.3994	-0.6106	-0.5735	-0.3897	-0.5713	-0.5070	-0.3861	-0.4626	-0.5154
	-0.9518			-1.0103			-0.9760		
1560	+0.1976	+0.2334	+0.2804	+0.2890	+0.1855	+0.1983	+0.1671	+0.1666	+0.2158
	+0.0916	+0.1064	+0.1416	+0.1663	+0.0759	+0.0810	+0.0615	+0.0584	+0.1006
	-0.0229	-0.0278	+0.0096	+0.0388	-0.0098	-0.0012	-0.0280	-0.0120	-0.0003
	-0.1889	-0.2062	-0.1630	-0.1444	-0.1482	-0.0978	-0.1502	-0.1262	-0.1132
	-0.4247	-0.6265	-0.6012	-0.4100	-0.5094	-0.4429	-0.3870	-0.4469	-0.4399
	-0.9928			-1.0410			-0.9729		
1575	+0.2151	+0.2068	+0.2482	+0.2242	+0.1822	+0.1434	+0.1314	+0.1231	+0.2286
	+0.0980	+0.0871	+0.1229	+0.1115	+0.0828	+0.0616	+0.0432	+0.0478	+0.1139
	-0.0190	-0.0477	-0.0005	-0.0293	-0.0124	-0.0049	-0.0425	-0.0126	+0.0117
	-0.1600	-0.2112	-0.1666	-0.2570	-0.1460	-0.0908	-0.1660	-0.0849	-0.1295
	-0.4318	-0.6378	-0.5537	-0.7535	-0.5029	-0.3567	-0.3679	-0.1853	-0.5037
	-1.0059						-0.9129	-0.5206	
1590	+0.2160	+0.1972	+0.2544	+0.1421	+0.1662	+0.1862	+0.1590	+0.1268	+0.1672
	+0.1129	+0.1048	+0.1263	+0.0588	+0.0622	+0.0893	+0.0567	+0.0575	+0.0727
	+0.0021	-0.0093	+0.0007	-0.0248	-0.0166	-0.0037	-0.0308	-0.0072	-0.0092
	-0.1261	-0.1398	-0.1723	-0.1340	-0.1373	-0.1010	-0.1659	-0.0799	-0.1035
	-0.2974	-0.3404	-0.5780	-0.3446	-0.4312	-0.3700	-0.3746	-0.1768	-0.3647
	-1.1682	-0.7734		-0.7468			-0.8638	-0.4654	
1605	+0.2051	+0.1873	+0.2252	+0.1828	+0.1746	+0.2068	+0.1403	+0.0983	+0.2206
	+0.1050	+0.1019	+0.1135	+0.0900	+0.0683	+0.0918	+0.0485	+0.0356	+0.1051
	-0.0073	-0.0072	-0.0125	-0.0066	-0.0186	-0.0015	-0.0300	-0.0132	-0.0012
	-0.1606	-0.1284	-0.1640	-0.1287	-0.1410	-0.1151	-0.1523	-0.0709	-0.1516
	-0.4342	-0.3246	-0.5337	-0.3486	-0.4691	-0.4366	-0.3457	-0.1612	-0.4900
	-1.0249	-0.7560		-0.8480			-0.7810	-0.4143	

N=50,000, PRF = 2KHz, Pulse Width = 200ns, Start Range = 1500m, Range Sample Rate = 10MHz, Pol = Horizontal